

entire program's code. For this reason, you should limit your use of global variables.

For example, let's look at the `calculateSlicesPerGuest()` function in a fictional *partyPlanner.py* program that is thousands of lines long. I've included line numbers to give you a sense of the program's size:

```
1504. def calculateSlicesPerGuest(numberOfCakeSlices):
1505.     global numberOfPartyGuests
1506.     return numberOfCakeSlices / numberOfPartyGuests
```

Let's say when we run this program, we encounter the following exception:

```
Traceback (most recent call last):
  File "partyPlanner.py", line 1898, in <module>
    print(calculateSlicesPerGuest(42))
  File "partyPlanner.py", line 1506, in calculateSlicesPerGuest
    return numberOfCakeSlices / numberOfPartyGuests
ZeroDivisionError: division by zero
```

The program has a zero divide error, caused by the line `return numberOfCakeSlices / numberOfPartyGuests`. The `numberOfPartyGuests` variable must be set to 0 to have caused this, but where did `numberOfPartyGuests` get assigned this value? Because it's a global variable, it could have happened anywhere in the thousands of lines in this program! From the traceback information, we know that `calculateSlicesPerGuest()` was called on line 1898 of our fictional program. If we looked at line 1898, we could find out what argument was passed for the `numberOfCakeSlices` parameter. But the `numberOfPartyGuests` global variable could have been set at any time before the function call.

Note that global constants aren't considered poor programming practice. Because their values never change, they don't introduce complexity into the code the way other global variables do. When programmers mention that "global variables are bad," they aren't referring to constant variables.

Global variables broaden the amount of debugging needed to find where an exception-causing value could have been set. This makes abundant use of global variables a bad idea. But the idea that *all* global variables are bad is a code smell myth. Global variables can be useful in smaller programs or